

Use Case Diagram & Documentation

Task 1 — Software Development Internship

Application: Weather Dashboard System

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1. Introduction and Purpose

This document presents the use case analysis for Task 1 of the CodeOrbit Tech Software Development Internship. The purpose of this task is to identify the actors and interactions of a small software system, represent them visually using a UML use case diagram, and document each use case with enough detail that a developer could implement it without ambiguity.

A use case diagram captures who uses a system (the actors) and what they can do with it (the use cases), without describing internal implementation. It forms the foundation for later design activities such as class diagrams, sequence diagrams, and sprint planning.

For this task, a Weather Dashboard System was selected as the sample application. It lets end users check current conditions and forecasts for any location, save favorite locations, and receive severe weather alerts, while an administrator manages the underlying data source and alert configuration. This scope is small enough to model completely within a one-month internship task, while still involving multiple actor types, an external system integration, and configurable business rules.

2. System Overview and Scope

The Weather Dashboard System is a web/mobile application that displays current weather conditions and forecasts for user-selected locations, sourced from a third-party weather data provider. Users can save favorite locations for quick access and configure alerts for severe weather events. Administrators configure the data source connection and the thresholds that trigger alerts.

The system is scoped to current conditions and short-range forecasting only. It does not include historical weather archives, minute-by-minute live radar tracking, or support for multiple simultaneous third-party data providers beyond a configured primary and fallback source. These exclusions are intentional simplifications appropriate for a one-month learning exercise and are restated formally in Section 7.

At a high level, the system supports three categories of functionality:

- Weather discovery and personalization, used by end users
- Alerting, triggered by system rules and delivered to users
- Configuration and data source management, used by admins

3. Actor Identification and Descriptions

Three actors were identified for this system, each with a distinct set of responsibilities.

3.1 User

The User is any end user of the dashboard, whether accessing it via web or mobile. Users search for locations, view current weather and forecasts, save favorite locations, configure their own alert preferences, and receive notifications when severe weather is detected for a saved location.

3.2 Admin

The Admin is a system operator responsible for keeping the dashboard's data pipeline and alerting rules accurate. Admins configure the connection to the external weather data

provider (API keys, refresh intervals, fallback sources) and define the system-wide thresholds that determine when a severe weather alert is triggered.

3.3 Weather Data Provider (external system)

The Weather Data Provider is an external third-party service (for example, a commercial weather API) that supplies raw current-condition and forecast data. It is modeled as an actor because it is an external system the application depends on, rather than a component the team is building — a common and correct use of actors in UML.

4. Use Case Diagram

The diagram below shows all three actors and their associated use cases within the system boundary. User-facing use cases are shown in purple, admin-facing use cases in coral, and the external data-provider use case in teal. A dashed <<include>> relationship shows that viewing current weather always involves retrieving data from the provider.

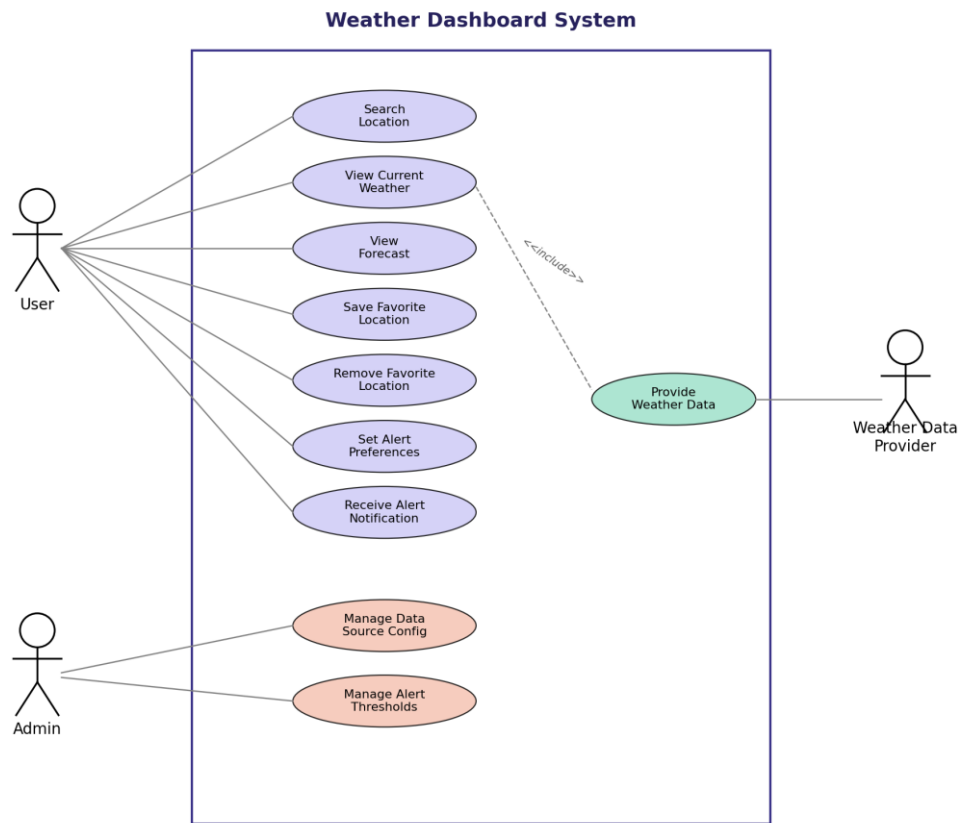


Figure : Use case diagram for the Weather Dashboard System.

5. Detailed Use Case Descriptions

Each use case identified in the diagram is documented below using a consistent template: actor, preconditions, main flow, postconditions, alternate/exception flows, and relevant business rules.

5.1 Search Location

Actor	User
Description	Allows a user to find a location by entering a city name or postal/zip code.
Preconditions	None — search is available without login.
Main flow	1. User enters a city name or zip code. 2. System queries the location database. 3. System displays matching results. 4. User selects a location to view its weather.
Postconditions	A location is selected and passed to the weather display.
Alternate flows	If no matches are found, the system displays a "location not found" message and suggests checking spelling.
Business rules	Search is limited to city-level granularity; exact GPS coordinate lookup is out of scope.

5.2 View Current Weather

Actor	User
Description	Displays real-time conditions — temperature, humidity, wind speed, and general conditions — for a selected location.
Preconditions	A location has been selected via search or chosen from favorites.
Main flow	1. User selects a location. 2. System requests current data from the Weather Data Provider (see 5.10). 3. System renders temperature, humidity, wind, and condition icon.
Postconditions	User sees up-to-date current conditions for the location.
Alternate flows	If the data provider is unreachable, the system displays the last cached reading along with a "data may be outdated" notice.
Business rules	Data is refreshed on a 15–30 minute cycle; true real-time (sub-minute) updates are out of scope.

5.3 View Forecast

Actor	User
Description	Displays an extended forecast (e.g., 5-day or hourly) for a selected location.
Preconditions	A location has been selected.
Main flow	1. User opens the forecast view for a location. 2. System retrieves forecast data from the provider. 3. System displays predicted temperature ranges and conditions per day/hour.
Postconditions	User sees the forecast for the selected location.
Alternate flows	If forecast data is temporarily unavailable, the system shows current conditions only with a message explaining the limitation.
Business rules	Forecast horizon is limited to 5 days; long-range or seasonal forecasting is out of scope.

5.4 Save Favorite Location

Actor	User
Description	Allows a user to bookmark a location for quick future access.
Preconditions	User is logged in; location has been searched or selected.
Main flow	1. User selects "Add to Favorites" on a location. 2. System adds the location to the user's favorites list.
Postconditions	The location appears in the user's favorites list on subsequent visits.
Alternate flows	If the user has already reached the maximum number of favorites, the system prompts them to remove one before adding a new one.
Business rules	Maximum of 5 favorite locations per user.

5.5 Remove Favourite Location

Actor	User
Description	Allows a user to remove a previously saved favorite location.
Preconditions	User has at least one saved favorite.
Main flow	1. User opens their favorites list. 2. User selects a location and confirms removal. 3. System removes it from the list.
Postconditions	The location no longer appears in the user's favorites.
Alternate flows	None.
Business rules	Removing a favorite also removes any alert preferences tied specifically to that location.

5.6 Set Weather Alert Preferences

Actor	User
Description	Allows a user to configure which severe weather alert types they want to receive, and for which locations.
Preconditions	User is logged in; at least one favorite location exists.
Main flow	1. User opens alert settings. 2. User selects alert types (e.g., storms, extreme heat, flooding) per favorite location. 3. System saves preferences.
Postconditions	User's alert preferences are stored and used to filter future notifications.
Alternate flows	None.
Business rules	Alert types offered to users are drawn from the categories defined by the Admin in Section 5.9.

5.7 Receive Alert Notification

Actor	User
Description	Delivers a notification to the user when a severe weather event matches their configured preferences for a saved location.
Preconditions	User has set at least one alert preference; a matching severe weather condition has been detected.
Main flow	1. System detects incoming weather data exceeding an admin-defined threshold. 2. System checks which users have matching alert preferences for the affected location. 3. System sends a push/email notification to each matching user.
Postconditions	Affected users receive a timely alert.

Alternate flows	If notification delivery fails (e.g., no connectivity), the system retries a limited number of times before logging the failure.
Business rules	Alerts are only sent for the user's saved locations, not arbitrary searched locations.

5.8 Manage Data Source Configuration

Actor	Admin
Description	Allows an admin to configure the connection to the external weather data provider.
Preconditions	Admin is logged in with configuration permissions.
Main flow	1. Admin opens data source settings. 2. Admin enters/updates API keys, refresh interval, or fallback provider details. 3. System validates and saves the configuration.
Postconditions	The system uses the updated configuration for subsequent data requests.
Alternate flows	If validation fails (e.g., invalid API key), the system rejects the change and displays an error.
Business rules	A fallback provider may be configured to be used automatically if the primary provider is unreachable.

5.9 Manage Alert Thresholds

Actor	Admin
Description	Allows an admin to define or adjust the system-wide thresholds that determine when a severe weather alert is triggered.
Preconditions	Admin is logged in with configuration permissions.
Main flow	1. Admin opens threshold settings. 2. Admin sets or edits values (e.g., wind speed above 60 km/h, temperature above 40°C). 3. System saves the updated thresholds.
Postconditions	New thresholds are applied to future incoming weather data.
Alternate flows	None.
Business rules	Threshold changes apply system-wide and affect all users' alert evaluations going forward.

5.10 Provide Weather Data

Actor	Weather Data Provider (external system)
Description	Supplies raw current-condition and forecast data to the system on a scheduled basis.
Preconditions	A valid data source configuration exists (see 5.8).
Main flow	1. System requests data from the provider at the configured interval. 2. Provider returns current conditions and forecast data. 3. System parses and stores the data for display and alert evaluation.
Postconditions	The system's weather data is refreshed and available to users.
Alternate flows	If the primary provider fails to respond, the system automatically retries the configured fallback provider, if one exists.
Business rules	Data refresh interval is admin-configurable, defaulting to every 15–30 minutes.

6. Actor–Use Case Traceability Matrix

The table below cross-references each actor against the use cases they participate in, confirming full coverage of the system's functional scope.

Use case	User	Admin	Data Provider
Search Location	✓		
View Current Weather	✓		
View Forecast	✓		
Save Favorite Location	✓		
Remove Favorite Location	✓		
Set Alert Preferences	✓		
Receive Alert Notification	✓		
Manage Data Source Config		✓	
Manage Alert Thresholds		✓	
Provide Weather Data			✓

7. Assumptions and Constraints

- The system supports only current conditions and a 5-day forecast; historical weather archives are out of scope.
- Weather data refreshes on a 15–30 minute cycle; sub-minute live updates are not modeled.
- A user can save a maximum of 5 favorite locations.
- Location search operates at city/zip-code granularity, not exact GPS coordinates.
- The system relies on one primary third-party weather API, with an optional single configured fallback provider.
- Alert categories available to users (e.g., storms, extreme heat, flooding) are defined centrally by the Admin, not created freely by users.
- Notification delivery (push/email) is assumed to use an existing notification service; building that service from scratch is out of scope.
- The system assumes a single admin role; there is no multi-tier admin permission model.

8. Non-Functional Considerations

Although use case diagrams primarily capture functional requirements, a few non-functional expectations were noted during analysis, as they influence how use cases should be implemented:

- Location search should return results within 2 seconds.
- The system should gracefully degrade to cached data if the weather provider is temporarily unreachable, rather than showing an error to the user.
- Alert evaluation should run promptly after new data arrives so that notifications remain timely for genuinely severe conditions.

9. Glossary of Terms

Actor	A user or external system that interacts with the system to achieve a goal.
Use case	A discrete unit of functionality that provides value to an actor.
Favorite location	A location a user has bookmarked for quick repeated access.
Threshold	A configured numeric limit (e.g., wind speed, temperature) that, when exceeded, triggers an alert.
Fallback provider	A secondary weather data source used automatically if the primary source is unavailable.
Precondition	A condition that must be true before a use case can begin.
Postcondition	A condition guaranteed to be true after a use case completes successfully.

10. Conclusion

This document has identified the three actors of the Weather Dashboard System — User, Admin, and the external Weather Data Provider — and documented ten use cases spanning location search, current conditions, forecasting, favorites management, alerting, and administrative configuration. Together with the use case diagram in Section 4, this analysis provides a complete functional baseline suitable for handing off to a development team, and it satisfies the requirements of Task 1 of the internship.